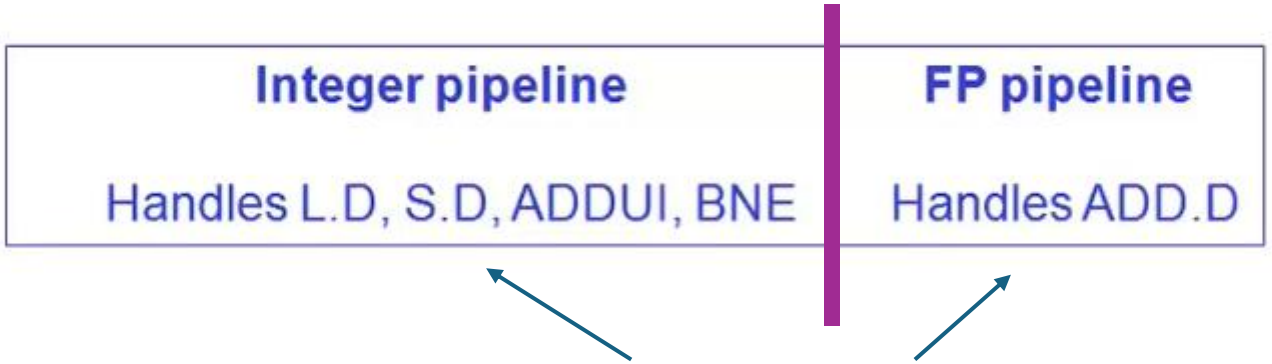
VLIW Example 1

VLIW Example 1

CS6810 -- Lecture 17. Lectures on Compiler-Based ILP.

https://www.youtube.com/watch?v=Q45iEIq3V3w&list=PL8EC1756A7B1764F6&index=17

Superscalar Pipelines



Two pipelines are available

- What is the schedule with an unroll degree of 4?

Superscalar Pipelines

5 unrolled loop example

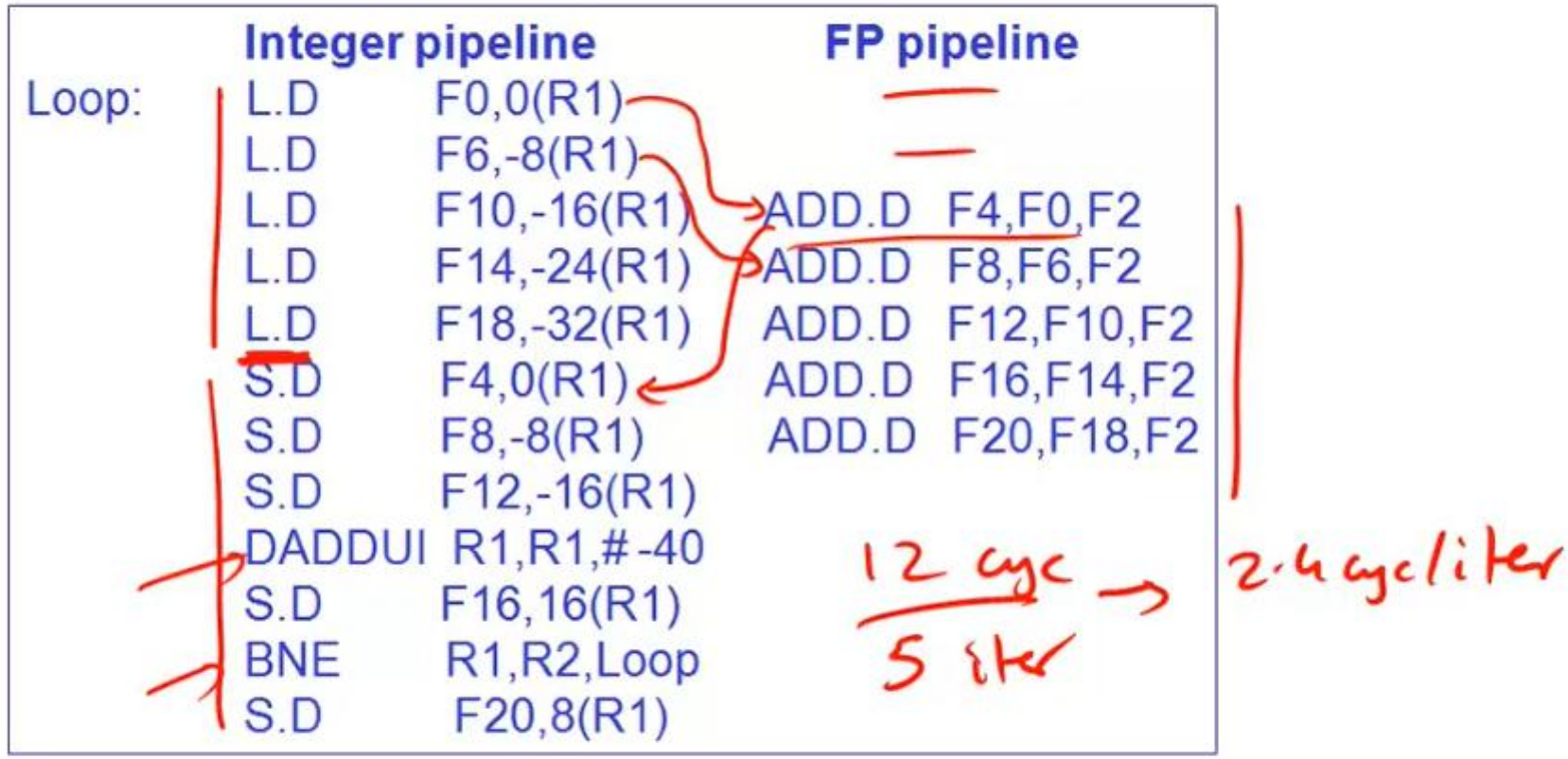
Similar to starting code.
Difference is 5 unrolled loops.
Below is 4 unrolled loop.

```
Loop:  L.D     F0, 0(R1)
       L.D     F6, -8(R1)
       L.D     F10,-16(R1)
       L.D     F14, -24(R1)
       ADD.D   F4, F0, F2
       ADD.D   F8, F6, F2
       ADD.D   F12, F10, F2
       ADD.D   F16, F14, F2
       S.D     F4, 0(R1)
       S.D     F8, -8(R1)
       DADDUI  R1, R1, # -32
       S.D     F12, 16(R1)
       BNE     R1,R2, Loop
       S.D     F16, 8(R1)
```

	Integer pipeline		FP pipeline	
Loop:	L.D	F0,0(R1)		
	L.D	F6,-8(R1)		
	L.D	F10,-16(R1)	ADD.D	F4,F0,F2
	L.D	F14,-24(R1)	ADD.D	F8,F6,F2
	L.D	F18,-32(R1)	ADD.D	F12,F10,F2
	S.D	F4,0(R1)	ADD.D	F16,F14,F2
	S.D	F8,-8(R1)	ADD.D	F20,F18,F2
	S.D	F12,-16(R1)		
	DADDUI	R1,R1,#-40		
	S.D	F16,16(R1)		
	BNE	R1,R2,Loop		
	S.D	F20,8(R1)		

- Need unroll by degree 5 to eliminate stalls
- The compiler may specify instructions that can be issued as one packet
- The compiler may specify a fixed number of instructions in each packet: Very Large Instruction Word (VLIW)

Superscalar Pipelines



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Superscalar Pipelines



VLIW

	Integer pipeline		FP pipeline	
Loop:	L.D	F0,0(R1)		
	L.D	F6,-8(R1)		
	L.D	F10,-16(R1)	ADD.D	F4,F0,F2
	L.D	F14,-24(R1)	ADD.D	F8,F6,F2
	L.D	F18,-32(R1)	ADD.D	F12,F10,F2
	S.D	F4,0(R1)	ADD.D	F16,F14,F2
	S.D	F8,-8(R1)	ADD.D	F20,F18,F2
	S.D	F12,-16(R1)		
	DADDUI	R1,R1,#-40		
	S.D	F16,16(R1)		
	BNE	R1,R2,Loop		
	S.D	F20,8(R1)		

12 cyc → 5 iter

2.4 cycle/iter

- Need unroll by degree 5 to eliminate stalls
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